ADVANCED COMPUTER NETWORKS-PCS223

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ME - CSE1

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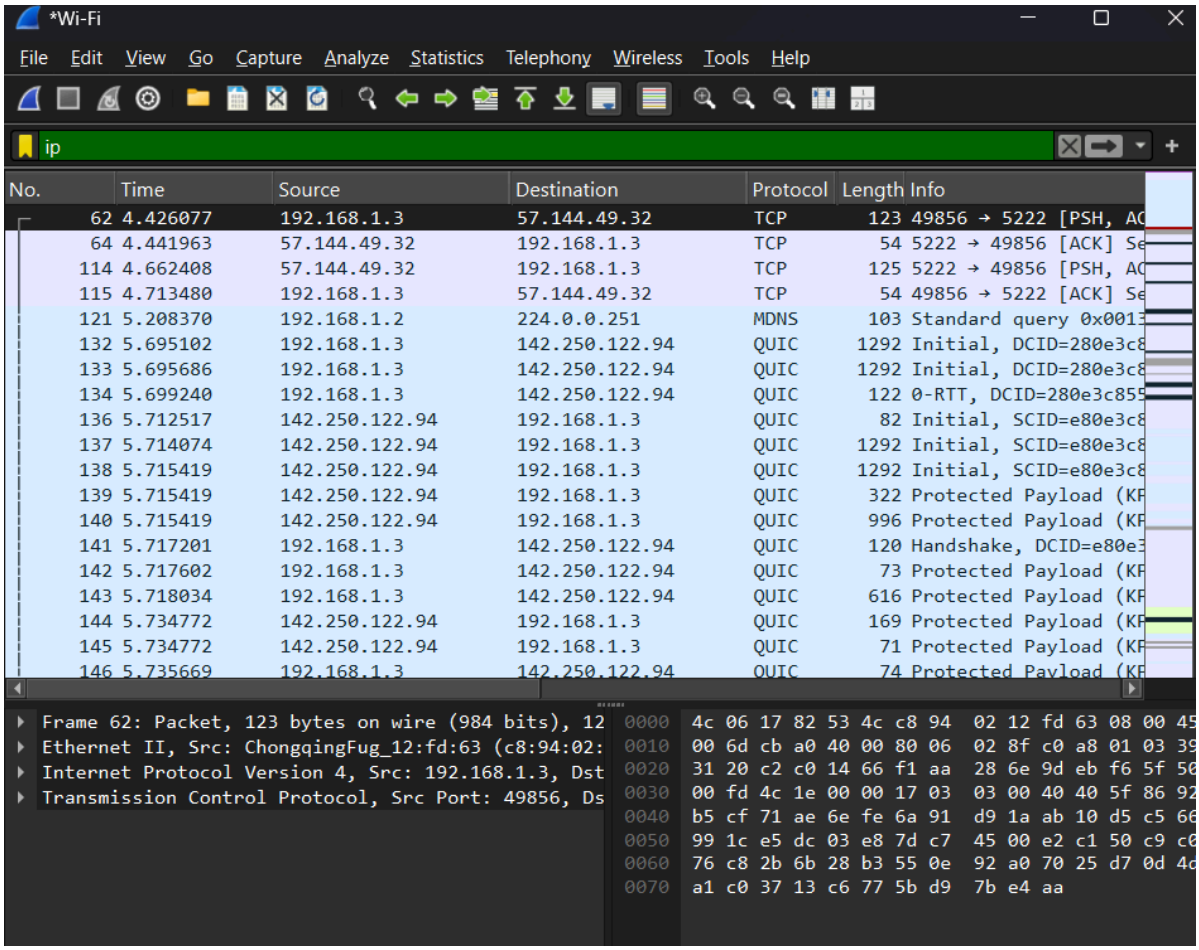
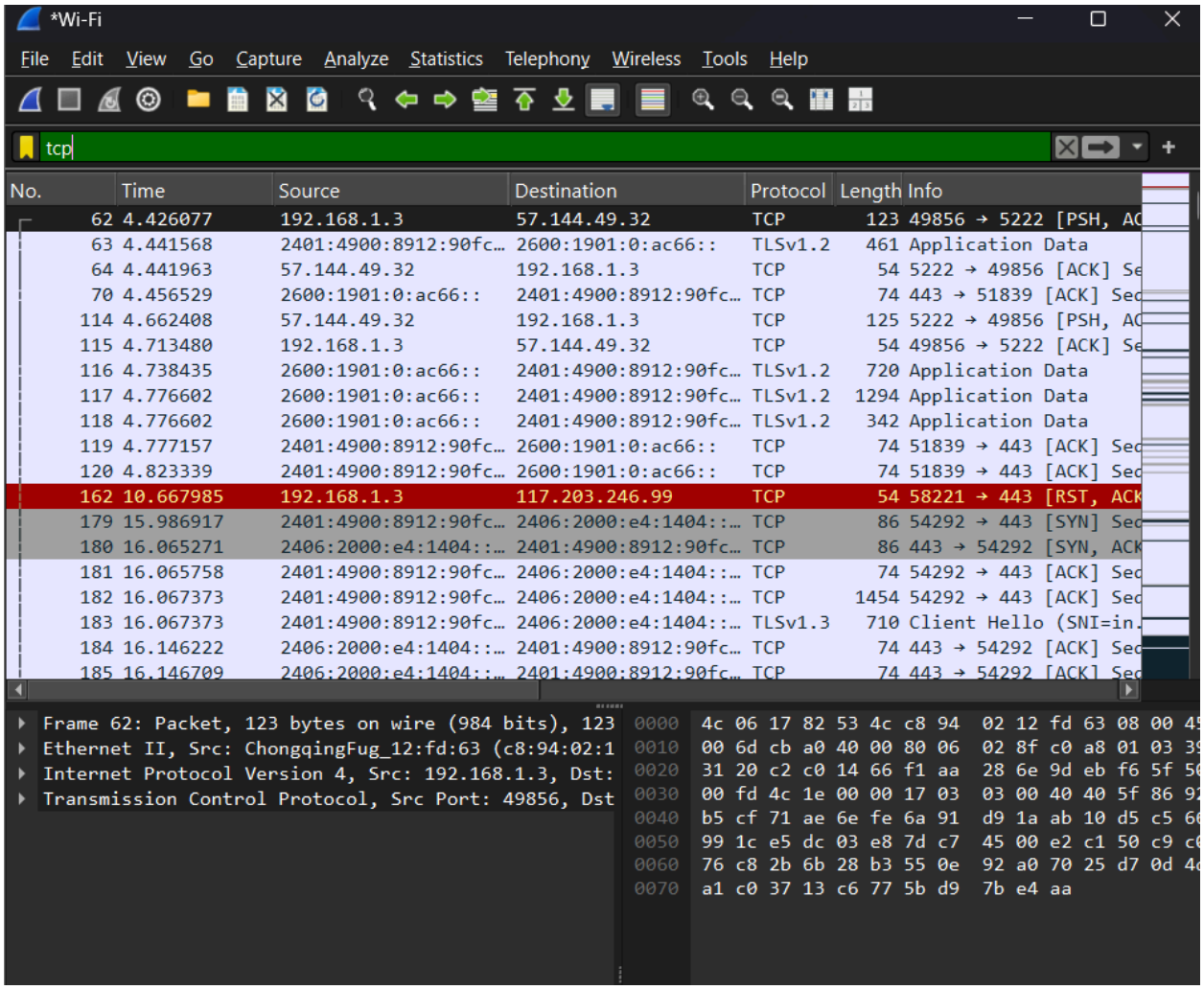
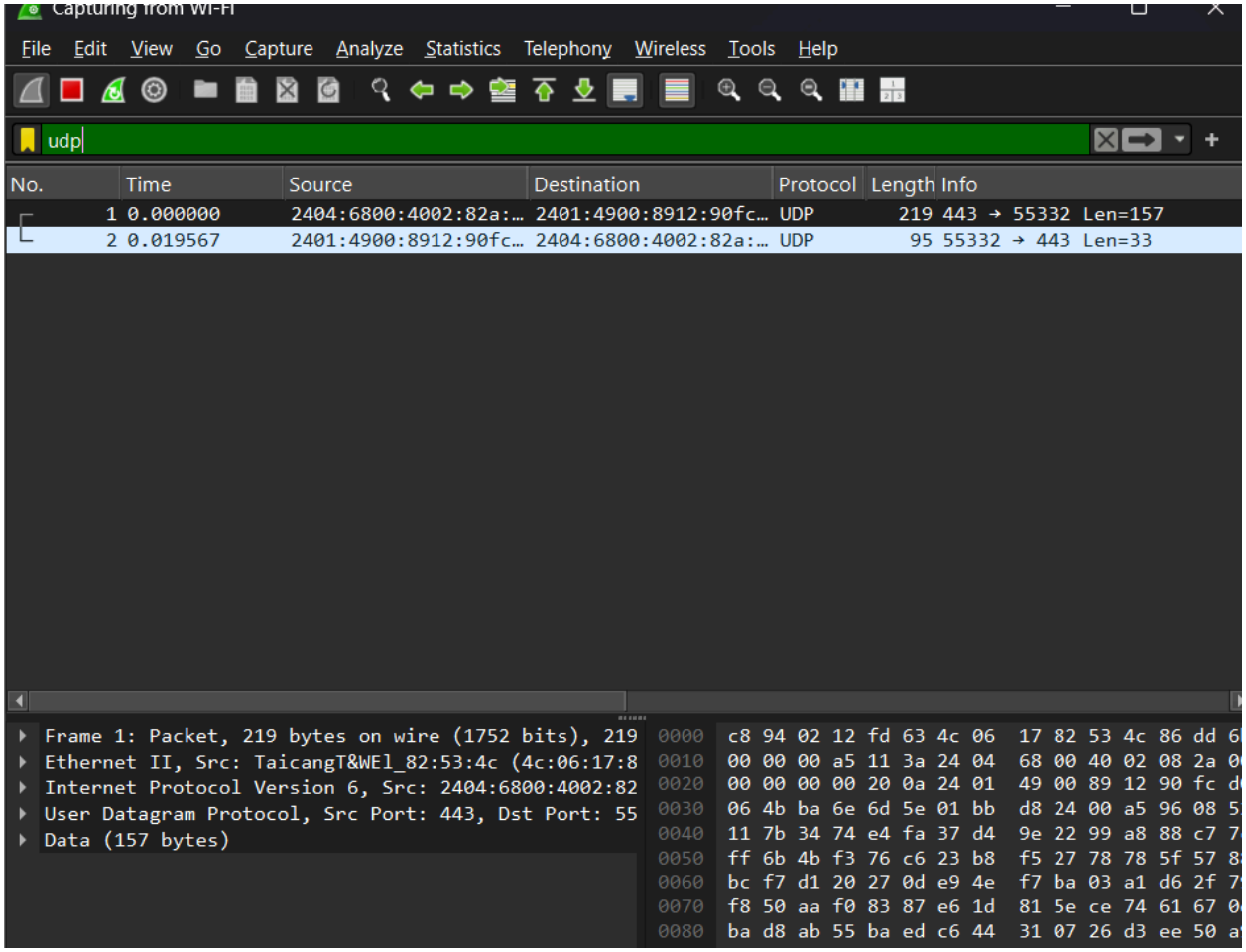
2026

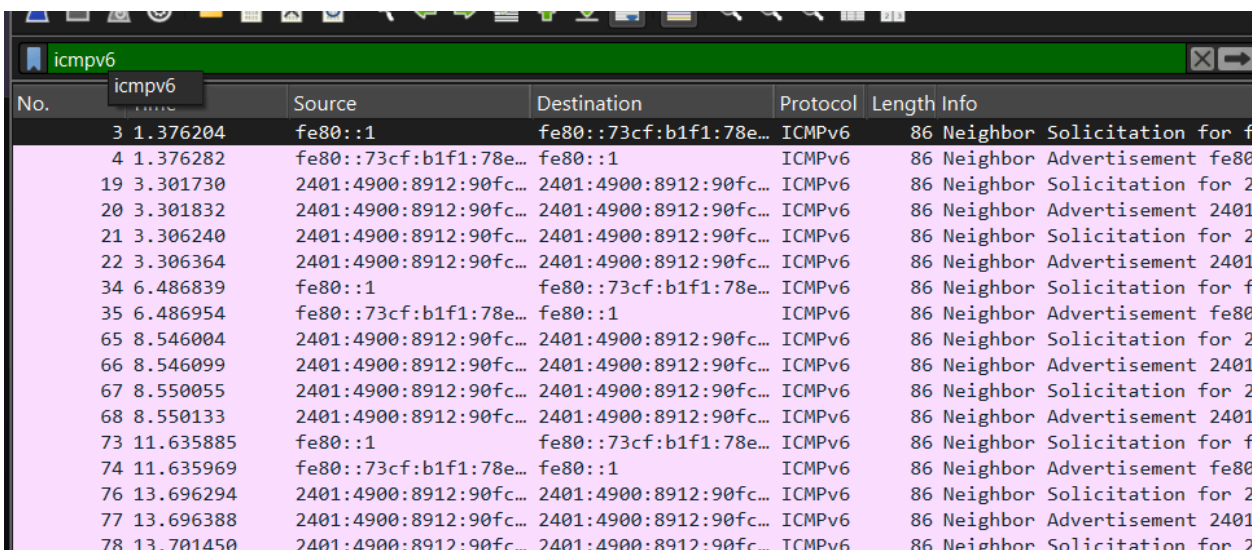
Assignment-1

1. To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

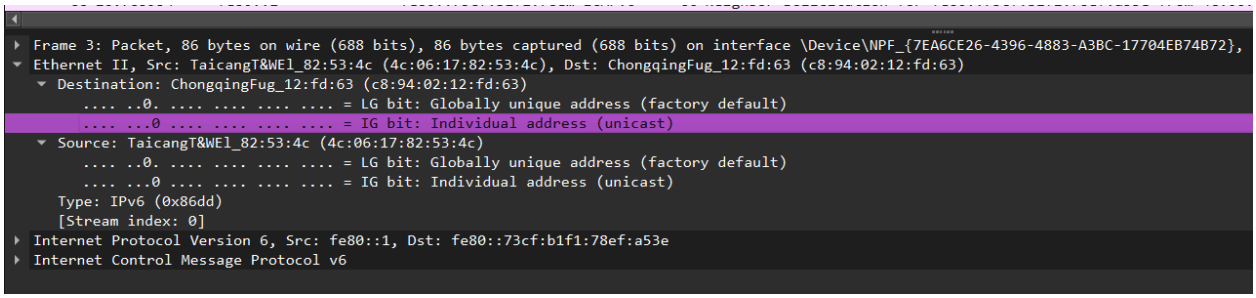



*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
62 4.426077 192.168.1.3 57.144.49.32 TCP 123 49856 → 5222 [PSH, AC
63 4.441568 2401:4900:8912:90fc… 2600:1901:0:ac66:: TLSv1.2 461 Application Data
64 4.441963 57.144.49.32 192.168.1.3 TCP 54 5222 → 49856 [ACK] Se
70 4.456529 2600:1901:0:ac66:: 2401:4900:8912:90fc… TCP 74 443 → 51839 [ACK] Sec
114 4.662408 57.144.49.32 192.168.1.3 TCP 125 5222 → 49856 [PSH, AC
115 4.713480 192.168.1.3 57.144.49.32 TCP 54 49856 → 5222 [ACK] Se
116 4.738435 2600:1901:0:ac66:: 2401:4900:8912:90fc… TLSv1.2 720 Application Data
117 4.776602 2600:1901:0:ac66:: 2401:4900:8912:90fc… TLSv1.2 1294 Application Data
118 4.776602 2600:1901:0:ac66:: 2401:4900:8912:90fc… TLSv1.2 342 Application Data
119 4.777157 2401:4900:8912:90fc… 2600:1901:0:ac66:: TCP 74 51839 → 443 [ACK] Sec
120 4.823339 2401:4900:8912:90fc… 2600:1901:0:ac66:: TCP 74 51839 → 443 [ACK] Sec
162 10.667985 192.168.1.3 117.203.246.99 TCP 54 58221 → 443 [RST, ACK
179 15.986917 2401:4900:8912:90fc… 2406:2000:e4:1404::… TCP 86 54292 → 443 [SYN] Sec
180 16.065271 2406:2000:e4:1404::… 2401:4900:8912:90fc… TCP 86 443 → 54292 [SYN, ACK
181 16.065758 2401:4900:8912:90fc… 2406:2000:e4:1404::… TCP 74 54292 → 443 [ACK] Sec
182 16.067373 2401:4900:8912:90fc… 2406:2000:e4:1404::… TCP 1454 54292 → 443 [ACK] Sec
183 16.067373 2401:4900:8912:90fc… 2406:2000:e4:1404::… TLSv1.3 710 Client Hello (SNI=in.
184 16.146222 2406:2000:e4:1404::… 2401:4900:8912:90fc… TCP 74 443 → 54292 [ACK] Sec
185 16.146709 2406:2000:e4:1404::… 2401:4900:8912:90fc… TCP 74 443 → 54292 [ACK] Sec
Frame 62: Packet, 123 bytes on wire (984 bits), 123
Ethernet II, Src: ChongqingFug_12:fd:63 (c8:94:02:1
Internet Protocol Version 4, Src: 192.168.1.3, Dst:
Transmission Control Protocol, Src Port: 49856, Dst
0000 4c 06 17 82 53 4c c8 94 02 12 fd 63 08 00 4
0010 00 6d cb a0 40 00 80 06 02 8f c0 a8 01 03 3
0020 31 20 c2 c0 14 66 f1 aa 28 6e 9d eb f6 5f 5
0030 00 fd 4c 1e 00 00 17 03 03 00 40 40 5f 86 9
0040 b5 cf 71 ae 6e fe 6a 91 d9 1a ab 10 d5 c5 6
0050 99 1c e5 dc 03 e8 7d c7 45 00 e2 c1 50 c9 c
0060 76 c8 2b 6b 28 b3 55 0e 92 a0 70 25 d7 0d 4
0070 a1 c0 37 13 c6 77 5b d9 7b e4 aa


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
1 0.000000 2404:6800:4002:82a:… 2401:4900:8912:90fc… UDP 219 443 → 55332 Len=157
2 0.019567 2401:4900:8912:90fc… 2404:6800:4002:82a:… UDP 95 55332 → 443 Len=33
Frame 1: Packet, 219 bytes on wire (1752 bits), 219
Ethernet II, Src: TaicangT&WEl_82:53:4c (4c:06:17:8
Internet Protocol Version 6, Src: 2404:6800:4002:82
User Datagram Protocol, Src Port: 443, Dst Port: 55
Data (157 bytes)
0000 c8 94 02 12 fd 63 4c 06 17 82 53 4c 86 dd
0010 00 00 00 a5 11 3a 24 04 68 00 40 02 08 2a
0020 00 00 00 00 20 0a 24 01 49 00 89 12 90 fc
0030 06 4b ba 6e 6d 5e 01 bb d8 24 00 a5 96 08
0040 11 7b 34 74 e4 fa 37 d4 9e 22 99 a8 88 c7
0050 ff 6b 4b f3 76 c6 23 b8 f5 27 78 78 5f 57
0060 bc f7 d1 20 27 0d e9 4e f7 ba 03 a1 d6 2f
0070 f8 50 aa f0 83 87 e6 1d 81 5e ce 74 61 67
0080 ba d8 ab 55 ba ed c6 44 31 07 26 d3 ee 50


icmpv6
icmpv6
No. Source Destination Protocol Length Info
3 1.376204 fe80::1 fe80::73cf:b1f1:78e… ICMPv6 86 Neighbor Solicitation for
4 1.376282 fe80::73cf:b1f1:78e… fe80::1 ICMPv6 86 Neighbor Advertisement fe80
19 3.301730 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Solicitation for 2
20 3.301832 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Advertisement 2401
21 3.306240 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Solicitation for 2
22 3.306364 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Advertisement 2401
34 6.486839 fe80::1 fe80::73cf:b1f1:78e… ICMPv6 86 Neighbor Solicitation for
35 6.486954 fe80::73cf:b1f1:78e… fe80::1 ICMPv6 86 Neighbor Advertisement fe80
65 8.546004 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Solicitation for 2
66 8.546099 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Advertisement 2401
67 8.550055 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Solicitation for 2
68 8.550133 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Advertisement 2401
73 11.635885 fe80::1 fe80::73cf:b1f1:78e… ICMPv6 86 Neighbor Solicitation for
74 11.635969 fe80::73cf:b1f1:78e… fe80::1 ICMPv6 86 Neighbor Advertisement fe80
76 13.696294 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Solicitation for 2
77 13.696388 2401:4900:8912:90fc… 2401:4900:8912:90fc… ICMPv6 86 Neighbor Advertisement 2401

OUTPUT-



| 1 | Frame Number | 3 |

| 2 | Packet Length | 86 bytes |

| 3 | Source MAC Address | 4c:06:17:82:53:4c |

| 4 | Destination MAC Address | c8:94:02:12:fd:63 |

| 5 | Source MAC Manufacturer | TaicangT&WE1 |

| 6 | Destination MAC Manufacturer | ChongqingFug |

| 7 | Ethernet Type | 0x86dd (IPv6) |

| 8 | IP Version | IPv6 |

| 9 | Source IP Address | fe80::1 |

| 10 | Destination IP Address | fe80::73cf:b1f1:78ef:a53e |

| 11 | IP Address Type | Link-local (fe80::/10) |

| 12 | Transport Protocol | ICMPv6 |

| 13 | LG Bit (Source) | 0 (Globally unique) |

| 14 | IG Bit (Source) | 0 (Unicast) |

| 15 | Capture Interface | NPF_{7EA6CE26-4396-4883-A3BC-17704EB74B72} |